

Color Dependent K-Means Clustering for Color Image Segmentation of Colored Medical Images

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Abstract—In the present work, a neoteric image segmentation technique has been framed, which is stood on color of the image using an unsupervised K-means clustering. The color image is converted into Lab (L=luminosity layer; a=chromaticity layer 1; b = chromaticity layer 2) in various computational steps and each layer has its own importance. Clustering is a process to distinguish different kind of objects in an image and K Means clustering partitions the image, such that within each cluster same type of objects are as close as possible and each cluster must be distinguished. Having several clusters we segment the nuclei into a separate image by recalling L layer. The proposed technique enables detection and analysis of objects without feature computation of every pixel in the image. Support vector machine (SVM) classified the disease by comparing the selected clustered image which is very much close for detection of disease with the existing standard data. The results so obtained with the proposed technique have been shown, which clearly depicts the segmentation of the complex colored medical images.

Keywords: Image segmentation, K-means clustering, Lab color space, Spatial resolution, Ultrasound image, Support Vector Machine (SVM).

I. INTRODUCTION

The process of splitting an image into some regions stood on some algorithms is called as segmentation [1], [2], [3], [4], [5]. Various significances of image segmentation are: content based-image retrieval, face recognition, face detection, object detection, fingerprint recognition, medical imaging, machine vision, iris recognition and video surveillance, etc[6]. Discontinuity and similarity are the two basic qualities of intensity values which cause all the algorithms of image segmentation. If the image is partitioned according to the abrupt changes in intensity then this process of partitioning comes under discontinuity, and if image is partitioned into some defined regions then this process of partitioning comes under similarity. Based on these properties image segmentation is basically of five types: Pixel based segmentation [7], Region based segmentation [8], Edge based segmentation, Edge and region hybrid based segmentation and clustering based segmentation [9]. All these segmentation works well with gray scale images but they have limitations with color images. Color image segmentation with $L^*a^*b^*$ color space based on genetic algorithm [10], extracts the object in a set of contour which is only based on objects present in the image. Hence, in this technique we have to analyze the feature of each pixel of the image, Which becomes computationally very difficult and consumes more time. The proposed segmentation technique is based on discontinuity property of intensity of color.

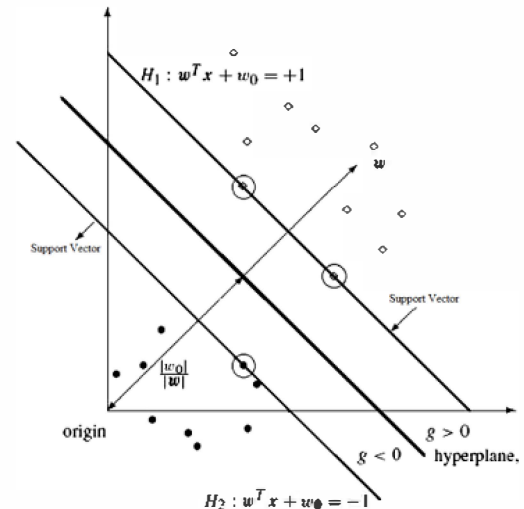


Fig. 1. 1:2-class hyperplane classifier

In medical images, acquisition and analysis of images is very difficult in order to detect the disease. Color based segmentation [10] is technique in which a colored image is partitioned into several clusters based on color. Here based on color means pixels of same features are separated together in a cluster. In the proposed method segmentation is based on an iterative method which segments the image into a number of clusters as per required analysis. After successful segmentation of the colored brain images which results in several clusters, we use SVM classifier to detect the existence of tumor in the image by comparing the suitable clustered image with the various data.

II. SUPPORT VECTOR MACHINE

Support vector machine (SVM) is most successful classifier. It classifies the data based on statistical pattern [11]. In some cases SVM is also called as a binary classifier in which, according to hyperplane, data is classified. The data is divided into classes with the aid of hyperplane in such a manner so that maximum margin should be there between two classes as shown in the following figure 1.

In this way, high dimensional space is classified by using Nonlinear data. Classification of data is done by using a kernel. There are many types of kernels are used to classify the data. Among these, polynomial functions and Gaussian radial-basis

functions are most common [12]. Lets assume

$$a = ai, i = 1 \dots n \quad (1)$$

and two classes are there which are ,1, 2 then the function will be written as

$$g(a) = W^T a + W_0 \quad (2)$$

The classification criteria for the data classification for fig 1 is given as:

if $g(a) > 0$ then $a \in 1$ and $b_i = +1$

$$if g(a) < 0 then a \in 1 and b_i = 1 \quad (3)$$

where $b_i=1$, $i=1 \dots n$ are class indicator values. SVM has the property of multi-classification which is done into two way [13]: first one is one-against-one and second one is one-against-all. In one-against-one case, for K data, there will be requirement of $K(K-1)/2$ SVM classifiers and one class data is distinguished from others by each classifier. In case of one-against-all, for K classes, K SVM classifiers will be used and here one class data is separated from K-1 classes.

The paper is arranged in the following manner. Section II gives the brief discussion about SVM. Section III gives the brief description of the methodology of the paper which includes K-means clustering and proposed algorithm with the help of flow chart. Results are discussed in section IV with the help of a brain image and the paper is concluded in section V.

III. METHODOLOGY

In the following Fig. 2, a flow chart is given which consists the steps which we have used in our methodology.

For better image segmentation, object should be in dark front and background should be light, so that the object can be easily separated from the background through thresholding [14].

Conditions for segmentation of a color image into some clusters are as follows:

- 1) Each cluster should be homogeneous and uniform with respect to some features.
- 2) In each cluster, same type of pixels should be as close as possible and every cluster should be as far as possible from others.
- 3) Each segment should have simple and spatially accurate boundaries.
- 4) Each segment should have simple interior and there should not be many small holes.

The process in which an image is partitioned into k groups automatically is called as K-means clustering [15]. In this process initially k cluster centers are selected then iteratively refined as follows: 1) The closest cluster centers are assigned instances. 2) Center of Each cluster is assumed to be the mean of its constituent instances.

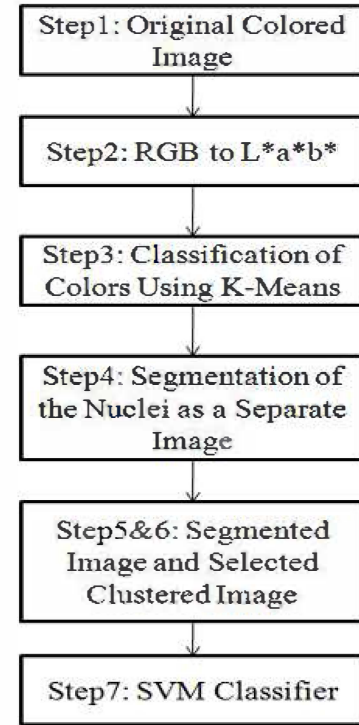


Fig. 2. Flow chart of the proposed algorithm

This algorithm will converse if assignment of cluster instances is not changing further. Each cluster is defined by its centroid or center. The point to which distances from all the objects are added to minimize is known as centroid. The outcome is a set of clusters which are as close and well-segregated as possible. After applying clustering algorithm to an image, number of clusters depends on number of peaks of histogram, particularly in K-means clustering algorithm. There will be k image clusters for k image histogram peaks using K-means clustering algorithm. Image histogram peaks can be calculated by smoothing the image histogram and then derivation to curve fitting and getting curve peaks and according to the distance threshold we make a responsible choice. If the threshold distance is greater than the point near to the peak point then the point will be rejected and we get the peak point and similarly we get our cluster number.

A. Proposed Color based K-means clustering

In this paper the color based image segmentation of colored ultrasound images is performed in following steps:

- 1) Acquire Image: As given in Fig. 2, the first step is to read a color ultrasound image instead of gray image.
- 2) RGB to L*a*b* Color Space Conversion: The L*a*b* color space is referred to as CIELAB or CIE L*a*b* which is used to differentiate the visual in equalities between the colors existing in an image. The L*a*b* color space contains CIE XYZ tristimulus values. The L*a*b* space is defined by three layers in which first One is luminosity layer L*, second one is chromaticity layer a* which defines the red-green axis along which color falls ,and last one is chromaticity-layer b* which defines the blue-yellow

axis along which the color falls. a^* and b^* layers consists of all the information of the colors. Euclidean distance metric technique can be used to differentiate the dissimilarities between two colors.

- 3) Classification of the Colors present in image in a^* b^* Space Using K-Means Clustering algorithm: In clustering different groups of similar objects are separated. In k-means clustering every object has a location in space. K-means clustering deals with the objects which have a location in space. How many clusters are to be partitioned is decided in K-means clustering and how much close are two objects with each other is calculated by distance metric. The a^* b^* layers consists of all the color information exists in the image. K-Means divides the objects into several clusters using the Euclidean distance metric.
- 4) Display Results of Nearest Neighbor Classification: An index is labeled by K-means with respect to a cluster, for each object. The label matrix consists of a color label for each image. This label matrix separates the objects in original image based on color.
- 5) Display a^* and b^* Values of the Colors: By plotting the a^* and b^* pixel values which are classified into separate colors, the different color populations is separated by the nearest neighbor classification. To display, each point should be labeled with its color label.
- 6) Segmentation of the Nuclei as a Separate Image: Each cluster consists of dark background and light front. L^* layer in the $L^*a^*b^*$ color space [16] performs a very important job to separate the dark background from light front.
- 7) Detection of tumor using Support Vector Machine (SVM) Classifier:

There are two types of classifiers: Linear classifiers and support vector machine. In linear classifiers, two methods are there: Linear separability and perceptron and In SVM classifiers: wide margin, cost function, slack variables, loss functions revisited, and optimization. Linear classifier can be expressed as

$$f(a) = WTX + c \quad (4)$$

For 2D, 3D and nD the discriminant is a line, a plane and a hyperplane respectively. In case of 2D, W is perpendicular to the line and is referred to as weight vector and c is the bias

And in case of 3D, a K-NN classifier required to carry the training data. Linear classifier uses the training data to learn W and then rejected; i.e. we need only W to classify new data. The another classifier known as perceptron classifier can be written as

$$f(ai) = WTAi + c \quad (5)$$

Which separates the categories for $i=1,2,3N$. SVM is an important part of linear classifiers and can be taken as next version of perceptron. These are the specific steps of Tikhonov regularization. They have a special property in which the empirical classification error and the geometric margin is minimized and maximized simultaneously respectively; hence both are maximum margin classifiers. SVM is compared with various classifiers [17]. SVM creates hyperplane or a

group of hyperplanes which is used to classify or to separate desired object by comparing the standard parameters. Using this concept we select that cluster which has the closest result. First we convert the selected color based cluster into grey scale then it is used to compare all the parameters with the standard data to get the information about whether the image is normal or abnormal for example if brain image has tumor then it is abnormal otherwise abnormal and this method gives the information about whether image has tumor or not.

IV. RESULTS AND DISCUSSIONS

The algorithm has been applied on a number of images. Experiments were accomplished on a large number of different kinds of images. The present investigation was done by considering three feature of image red, green and blue value of a pixel. To obtain the color based segmentation, colored images were taken. The proposed algorithm was tested with various standard color images like lena, baboon, house, boats, rivers, etc. The results for the standard colored images are shown in different experiments as follows:

A. Experiment 1: Lena

In following figure, fig. 3(a) is original image of lena and fig. 3(b-f) are the various segmented images with different clusters based on different colors present in the original image of lena.

B. Experiment 2: Baboon

In this experiment we have taken another standard colored image Baboon. 5 clusters based on different colors were obtained as shown in the following figures in which fig. 4(a) is the original image of Baboon and fig. 4(b-f) are the various segmented images with different clusters based on different colors present in the original image of Baboon.

C. Experiment 3: Brain

Satisfactory results were obtained with the use of proposed algorithm on standard colored images. Hence, this algorithm was also tested on colored medical images. Since, colored medical images are now used In medical fields for better visual results. This algorithm was tested on colored brain images and

Colored ultrasound images. Standard colored medical images were used for simulations. Fig. 5 shows various segmented parts of a colored brain image. From cluster 4 (fig. 5(e)) we can see that the tumor is clearly visible, which can be segmented out easily for further processing, but to decide that the resulted image has the tumor, we have to compare the image with the data which declare that whether image has tumor or not with the help of SVM classifier.

V. CONCLUSIONS

In this paper a modernistic method of color based segmentation of colored images has been proposed. As per observations, the proposed technique enables effective segment of the colored image iteratively based on the requirement. Using this methodology it is easy to visualize the results and analyze the particular object in the image. The previously existing

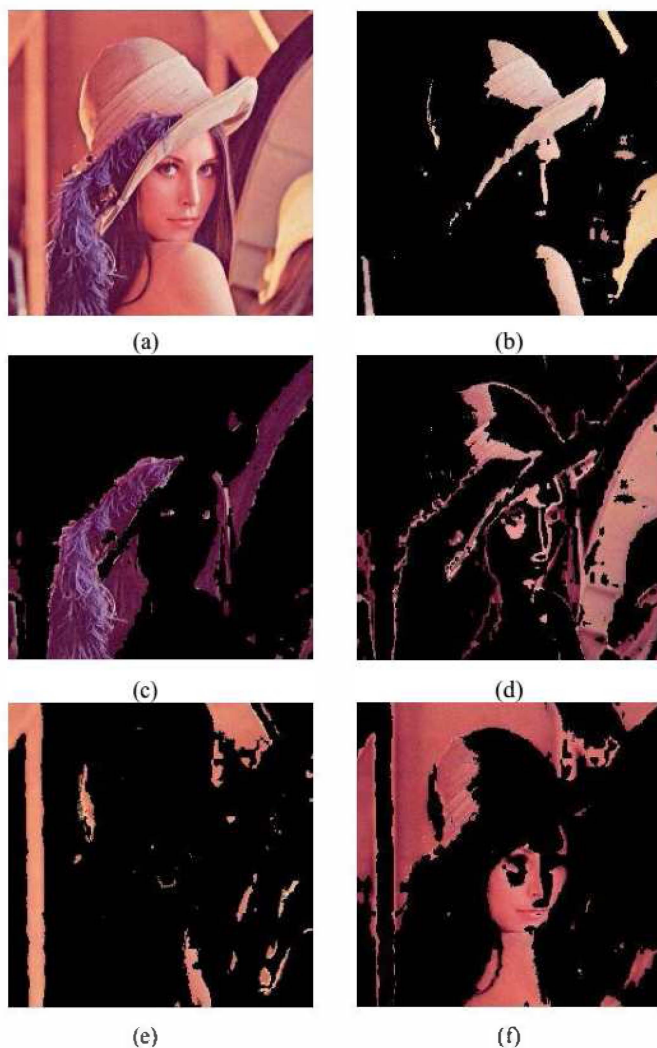


Fig. 3. Lena; (a) Original image, (b) Object in cluster 1, (c) Object in cluster 2, (d) Object in cluster 3, (e) Object in cluster 4, (f) Object in cluster 5

method provided segmentation only for gray scale images. The proposed technique is an effective and computationally

Fast tool towards segmentation of color image, which depends upon number of clusters in K-means clustering algorithm. The results can be enhanced by increasing cluster number. Since till now no colored based ultrasound brain images with information about the status of the brain whether it has tumor or not. So in future in some next years, on availability of colored ultrasound images we would be able to define the disease of the brain using this method. By using this method it can also be possible to examine the brain diseases other than tumor like blood clotting, swelling or other types of typical diseases and can also be beneficial for other part of the body whose ultrasound images can be taken.

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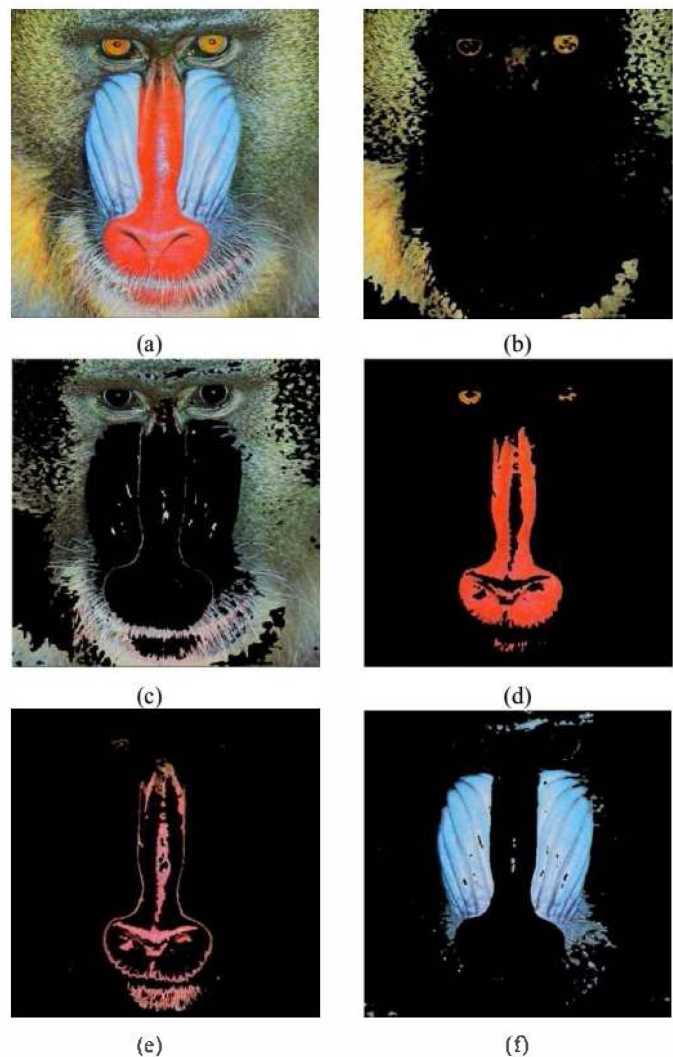


Fig. 4. Baboon; (a) Original image, (b) Object in cluster 1, (c) Object in cluster 2, (d) Object in cluster 3, (e) Object in cluster 4, (f) Object in cluster 5.

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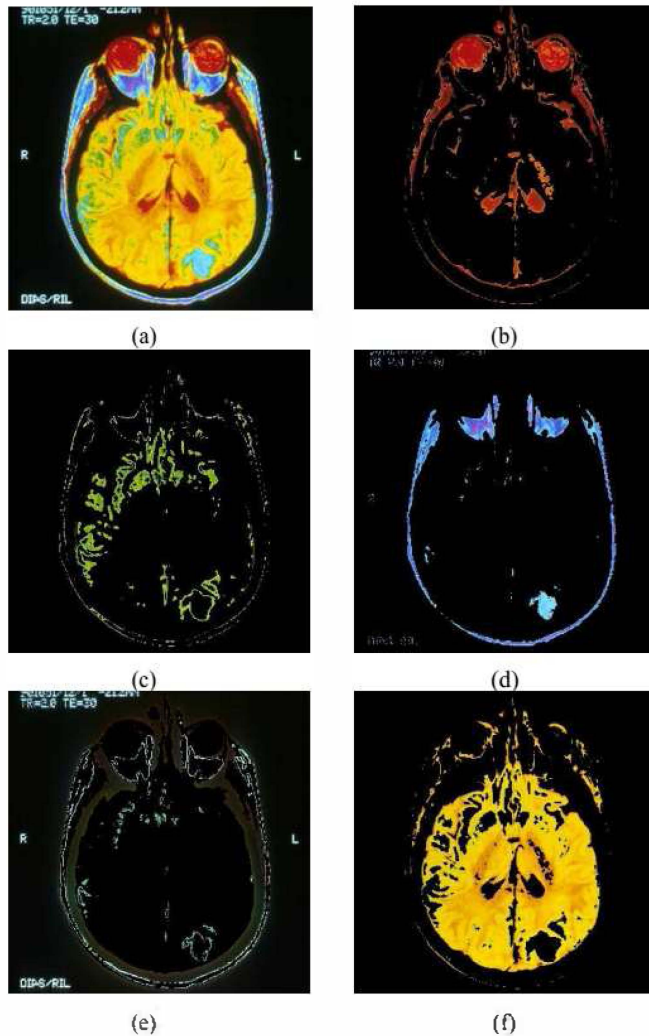


Fig. 5. Brain:(a)Original image, (b)Object in cluster1, (c)Object in cluster2, (d)Object in cluster3, (e)Object in cluster4, (f)Object in cluster5.

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